

Q. Write Kleene's theorem part I, Any RL can be accepted by a FA. (7 marks) OR Prove Kleene's theorem part I. (7 marks)

Kleene's Theorem states the equivalence of the following three statements –

- A language accepted by Finite Automata can also be accepted by a Transition graph
- A language accepted by a Transition graph can also be accepted by Regular Expression.
- A language accepted by Regular Expression can also be accepted by finite Automata.

Kleene's Theorem Part-1 : A language accepted by Finite Automata can also be accepted by a Transition graph.or, equivalently, by a **regular expression**.

Statement of Kleene's Theorem Part-I

If a language can be written using a **regular expression**, then we can always construct a **Finite Automaton (FA)** that accepts that language.

Proof of Kleene's Theorem Part-I

We prove this theorem using **structural induction** on regular expressions.

A regular language is built from:

1. Basic regular expressions
2. Operations:
 - Union
 - Concatenation
 - Kleene Star

Case 1: Language = $\emptyset$ (empty set)

This language accepts no strings.

A finite automaton can be made with:

- one start state
- no final state

So, no string is accepted.

Hence, $\emptyset$ can be accepted by FA.

Case 2: Language = $\{\epsilon\}$

This language contains only the empty string.

Construct FA:

- Start state is also final state.

So, ϵ is accepted.

Hence, $\{\epsilon\}$ can be accepted by FA.

Case 3: Language = $\{a\}$

Language contains single symbol a.

Construct FA:

- q_0 = start state
- q_1 = final state

Thus string a is accepted.

Hence, every single symbol language can be accepted by FA.

Since:

- basic regular expressions have finite automata
- and FA remains possible under:
 - union
 - concatenation
 - star operation

Therefore every regular language can be accepted by a finite automaton.

Hence proved.

A language is said to be a regular if it can be shown using a Finite Automaton (FA) or written as a Regular Expression (RE).

For simple expressions like $(a+b)$ or ab , we can easily draw the finite automaton. But for longer expressions, it becomes difficult.

To solve this, Kleene's Theorem gives a proper and systematic method to build finite automata.

- A regular language can be represented by FA or RE. Simple expressions are easy to convert into FA. Kleene's Theorem helps in converting complex expressions into FA.

